📘 SOC Internship – Day 6 Notes

Topic: SIEM (Security Information and Event Management)

🎯 Objective

Understand SIEM concepts, architecture, workflow, and real-world usage in SOC.

🧠 1. What is SIEM?

SIEM is a security platform that:

- Collects logs
- Correlates data
- Analyzes events in real-time

👉 Purpose: **Detect security threats efficiently**

⚙️ 2. Components of SIEM

1. Log Collection

- Collects logs from:
  - Servers
  - Firewalls
  - Endpoints

2. Log Storage

- Stores logs for future analysis

3. Correlation Engine

- Finds patterns across multiple logs

4. Alerting System

- Generates alerts for suspicious activities

5. Dashboard

- Visual interface for SOC analysts

🔄 3. How SIEM Works (Workflow)

Step-by-step process:

1. Collect logs from multiple systems
2. Normalize and store logs
3. Apply correlation rules
4. Detect suspicious activity
5. Generate alerts

👉 This process helps SOC analysts respond quickly to threats.

🌍 4. Real-World Example

- Multiple failed login attempts detected
- SIEM correlates events across systems
- Triggers **Brute Force Attack alert**

🛠 5. Popular SIEM Tools

- Splunk
- ELK Stack (Elasticsearch, Logstash, Kibana)
- IBM QRadar

6. SOC Use Cases

- Detect brute force attacks
- Identify malware activity
- Monitor insider threats
- Investigate security incidents

Assignment Answers

1. Explain SIEM Workflow

SIEM collects logs from different sources, stores them, analyzes patterns using correlation rules, detects suspicious activity, and generates alerts for SOC analysts.

2. List 3 SIEM Tools

- Splunk
- ELK Stack
- IBM QRadar

3. One Real-World Detection Example

Multiple failed login attempts from different systems are correlated, and SIEM detects it as a **brute force attack** and triggers an alert.

Key Takeaways

- SIEM = **central brain of SOC**
- Helps in **real-time threat detection**
- Uses **correlation rules to find hidden attacks**